Supplementary material for PLECTA: Overlap-aware instance reconstruction of dense strand networks from binary masks

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S1 Reconstruction: cost function, gates and schedule

This section gives the arithmetic behind Section 2.4 of the article. Values are those of the released configuration; nothing here was tuned on held-out or locked data.

S1.1 Skeleton graph

The mask M is binarised at > 0, skeletonised, and spurs no longer than 3 px are removed iteratively. Skeleton pixels of degree at least three form junction clusters, and adjacent clusters merge into nodes: connectors up to 5 px are absorbed into one node, connectors up to 14 px merge nearby nodes while remaining real arms, and free debris up to 2 px is absorbed. Each remaining connected path is stored as an ordered arm carrying one stub at each end.

S1.2 Stub frames

A stub frame holds its tip, outward tangent and signed curvature, estimated by a least-squares arclength polynomial fit read outward from the tip. The fit is quadratic when the span reaches 18 px and linear otherwise, and a frame counts as reliable only with at least three samples spanning at least 3 px. Local frames use 24 px of support; once links form chains that support expands to 55 px, so a stub too short to have a direction of its own borrows one from the chain it has joined.

S1.3 Weights and scales of the pairing cost

The cost function itself is stated in the article, which also gives the reasoning behind its three deliberate properties. What follows are the values it is evaluated with.

Junction and gap links use separate weight sets, because a junction link is scored on direction alone while a gap link must also answer for the distance it spans. The curvature term carries a constant of 30, which places curvature, in px^{-1}, on the scale of the angular terms; without it a term in reciprocal pixels and a term in radians could not be added. Junction links set $w_\ell = 0$, a junction having no gap to pay for, and ℓ_0 and d_0 set the length scale of the gap penalty and the width of the chord fade respectively.

S1.4 Penalties, gates and admissibility

Leaving a stub unmatched carries a penalty rather than a prohibition, and the penalty p_i also fixes admissibility: an edge is a candidate only when $C_{ij} < p_i + p_j$. Exact maximum-weight partial matching then maximises $\sum(p_i + p_j - C_{ij})$ over disjoint pairs, which minimises total link cost plus p_i for every stub left free.

Table S1. Cost weights and scales, read from the frozen configuration that produced every published number. Junction and gap links are scored with separate sets; the gap set alone carries a length term, a junction having no gap to pay for.

Symbol	Term	Junction	Gap
w_d	reversal angle θ	0.6	0.2
w_t	chord turns $\varphi_a + \varphi_b$	1.2	1
w_κ	curvature continuity	1	0.5
w_ℓ	length d/ℓ_0	—	0.15
ℓ_0	length scale (px)	—	60
d_0	chord-fade scale (px)	4	3
p_i	price of leaving a stub free	0.62	0.3

The admissibility limit is not a separate threshold. An edge with $C_{ij} \geq p_i + p_j$ contributes nothing positive to the objective, so a maximum-weight *partial* matching would never select it, and declining to offer it removes an edge that could not have been chosen.

At the configured penalties junction stubs carry $p_i = 0.62$, so a junction edge needs $C < 1.24$. Gap stubs carry $p_i = 0.30$, so a gap edge needs $C < 0.60$, and must in addition satisfy $d \leq 85$ px, $\theta \leq 0.40$ rad and $\varphi \leq 0.28$ rad at each end. The gap gates are the tighter set because a gap link asserts a continuation across evidence that is missing rather than merely ambiguous. The values were fixed on development scenes.

S1.5 The round schedule

Rounds $r = 0, \ldots, 8 - 1$ apply a schedule fraction $\alpha_r = 0.85 + (1 - 0.85)\, r/7$ that scales the unmatched penalties and the two gap angle limits, and nothing else. Scaling a penalty *down* admits fewer edges, admissibility being $C_{ij} < p_i + p_j$, and scaling an angle limit down narrows the gap gates, so early rounds are the conservative ones and are deliberately under-linked: while the frames are least reliable, only pairings that survive the reduced penalties and narrowed gates are committed, and the constraints relax towards their configured values as the chains assembled so far improve the frame estimates. The final round, $\alpha_7 = 1$, is the only one run at the configured parameters, and is the one returned.

In full:

1. Skeletonise M, prune spurs, merge junction clusters into nodes, and store the remaining paths as arms with a stub at each end.
2. For $r = 0, \ldots, 7$, re-solving both matchings from scratch and carrying only the chain structure forward from round $r - 1$:
   (a) Re-fit every stub frame along the chain it now belongs to.
   (b) At each node, form candidate edges between stubs of different arms with finite frames and cost, keep those admissible at α_r, and solve one maximum-weight matching per node.
   (c) From the still-unmatched reliable stubs, form gap candidates on different arms and different nodes that pass the distance gate and the α_r-scaled angle gates, and solve one global maximum-weight matching over the admissible ones.
3. Return the links of round $r = 7$.

S1.6 Why the junction is solved as a whole

Joint and sequential solutions coincide at plain crossings and diverge only as nodes grow dense. Over 20 development scenes they never disagreed at nodes of degree 2–4 ($n = 542$), disagreed at 22 % of nodes with 5–8 stubs ($n = 187$) and at 69 % of nodes with 9 or more ($n = 233$), where the joint solution was better by a median of 0.90 in total cost.

That comparison holds one node's costs fixed. Run inside the rounds, where each matching sets the frames the next is scored against, a cheap early choice propagates and the difference is no longer confined to dense nodes.

S1.7 Instance output

Connected components of accepted arm links define instances. PLECTA paints every arm pixel and every junction cluster touched by a chain into that instance, so intersecting instances share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the released JSON configuration.

S2 Depth stage: evidence and solver

This section gives the detail behind Section 2.6 of the article.

S2.1 Sampling geometry at a crossing

Where two instance centrelines intersect, a short arc of each is taken through the crossing (the shared *core*, a flat 7.5 px of arc, the same at every crossing), together with the *flanks* on either side of it. Flank samples are kept clear of the other strand's stroke, because a sample taken inside it measures the union of two rods rather than one of them. A crossing with too few usable flank samples abstains rather than being decided on thin evidence.

The core length was previously sized from the two strands' radii so that it spanned both silhouettes. It is now the constant above, and no per-rod radius enters the evidence path at all, so a radius estimate can no longer move a depth decision. Radii re-enter only downstream, in the non-interpenetration constraints that set metric height.

S2.2 One channel, and its noise floor

One quantity is read on core and flanks alike: median intensity. Write m_i and m_j for the two strands' flank medians and m_c for the pooled core median. The evidence at the crossing is the score

$$s \;=\; 2\,\frac{|m_c - m_j| - |m_c - m_i|}{\max(|m_i - m_j|,\; 2\hat{\sigma})}, \tag{S1}$$

whose sign names the upper strand, the one whose flanks the core resembles more closely, and whose magnitude $|s|$ is the weight that decision carries. A crossing abstains when $|s| < 0.40$; the threshold is set directly in score units rather than as a band around a probability. Here $\hat{\sigma}$ is a noise estimate belonging to that crossing alone: 1.4826 times the median absolute deviation of the within-rod flank residuals, each strand's flank samples being centred on their own median and the two residual sets pooled. It therefore measures the scatter within a rod's own appearance, not the difference between the two rods.

A probability p is still written into the per-crossing record so that the output keeps its schema, but it is a monotone relabelling of s. It is neither a calibrated confidence nor the quantity anything is decided or weighted on, and should not be read as one.

Why the denominator is a noise floor

By the reverse triangle inequality the numerator of (S1) is bounded in magnitude by $|m_i - m_j|$, so a denominator equal to that separation drives the ratio to ± 1 whenever the two strands differ at all. The previous form of this rule floored the denominator at a small fixed constant in grey levels, far below the noise of any of these images. The consequence was structural rather than statistical: a maximal score was then reachable on a separation smaller than the noise, and a maximal score exceeded any abstention threshold, so the rule could not abstain on such a crossing whatever the evidence actually was. Nor was that regime rare: a substantial fraction of crossings saturated this way, and among the saturated ones those whose separation fell below the noise were markedly less accurate than those separated by more than it. Those supporting measurements were made on the removed rule, which no configuration of the shipped code restores, so they are recorded outside this manuscript rather than reported here. Flooring at $2\hat{\sigma}$ instead makes a confident score mean *separated by more than the noise* rather than merely separated, and restores the stage's ability to abstain where it should.

The gradient channel, and why it was removed

An earlier version of this stage read a second quantity beside intensity: median gradient magnitude sampled on each strand's own two edges at centreline $\pm r\,\mathbf{n}$, the axis gradient being near zero and therefore uninformative. The argument was that the upper strand's edges stay crisp through the crossing while the lower strand's are overwritten. It did not survive measurement. Standing alone the channel decided almost no real crossings, agreed with the intensity channel at chance, and was itself at chance; those readings were taken on the removed channel, and no configuration of the shipped code reproduces them, so they are recorded outside this manuscript rather than reported here. The reason is structural, and is visible in the channel's own sampling geometry: within the core those $\pm r$ offsets fall inside the *other* strand's stroke, so the quantity is contaminated exactly where it would have had to be read. The channel is removed, and with it the gradient image it required, the two channel weights that combined it with intensity, and the logistic, since with one channel there is nothing to combine and s is already the decision variable.

What the replacement is worth

The two rules were compared on a held-out set of 20 optically rendered synthetic scenes carrying 1342 crossings, whose seeds are disjoint from every set used to develop either rule and to choose between them. Its constants are a weaker claim: the flat core half-size (7.5 px) was settled by a sweep run on this same set, and the abstention threshold's flatness was checked on it, so with respect to those two values the set is not held out. Neither was invented here – 7.5 px is exactly what the previous rule's radius-free fallback already computed, and both values sit on a plateau rather than a peak, a flat core being no worse than oracle per-rod radii anywhere over 6–14 px and the accuracy unchanged across a band of thresholds around 0.40, so nothing below turns on either value, but the comparison is in-sample with respect to them.

At matched coverage, meaning both rules made to answer as often as the shipped one does at 83.0 %, the one-channel rule matches the two-channel rule's accuracy: the difference is $+0.0045$, 95 % CI $[-0.0186, +0.0235]$, which is no difference, and no claim of higher accuracy is made. One reading does go against the replacement and is reported rather than omitted: forced to decide *every* crossing, including those it would decline, the one-channel rule is worse by -0.0142, CI $[-0.0313, +0.0007]$. That is the expected direction, because abstention is exactly where an ordered confidence pays, and removing it removes the advantage; it is also not how either rule is run. What improves is the ordering of the stage's own confidence. The area under the curve of correctness against $|s|$ rises from 0.6531 to 0.7586, a gain of $+0.1056$, CI $[+0.0659, +0.1425]$. That matters because $|s|$ is the edge weight the global ordering below is solved with: better-ordered confidence means contradictions are broken at the genuinely weakest relations rather than at arbitrary ones. It also converts into accuracy wherever more abstention is acceptable, by $+0.0054$ at 82.5 % coverage and $+0.0121$ at 80 %. The claim is therefore the same accuracy with better-ordered confidence, from a simpler evidence path.

A second channel, axial continuity along each strand, was built and measured on the same held-out set. It was inert, and is deliberately not included. That measurement was made on code that was never merged and cannot be rerun here, so it is recorded outside this manuscript rather than reported in it.

S2.3 One order for the whole field

Each decided crossing contributes a directed edge weighted by $|s|$ itself, oriented toward the relation the sign of s names. The score is the primitive: the reported p is a relabelling of it and enters no computation here. Within each connected component of the crossing graph one total order over its instances is derived so as to maximise the total satisfied weight, and every relation in the component is then re-read from that order: the ones the order disagrees with are flipped rather than discarded, and both the flip and the original local score are recorded, so a reader can see how much the global correction moved. Which relations that turns out to be is the subject of the paragraph below.

Maximising satisfied weight is the minimum-weight feedback arc set objective, but only components of at most 14 instances attain it, by dynamic programming over subsets; larger ones get a greedy order refined with adjacent swaps, which carries no optimality guarantee. Because the order is re-derived for the component as a whole rather than only where relations conflict, a greedily ordered component can return a relation reversed that nothing contradicts, which an exact solution never does; a reversal lying on no cycle only loses weight. Fed ground-truth crossings with their true directions, input that is acyclic by construction and that an exact solver would leave untouched, the pass still reverses 24 relations across the 20 held-out scenes, capping layer agreement at 0.886 where it should reach 1.000; a condense-then-order variant, which orders the condensation of the strongly connected components and reorders only inside genuine cycles, reverses none of them but measured worse on the actual noisy predictions and is not what ships.

S2.4 Layers and metric height

The corrected relations form a directed acyclic graph whose longest path fixes the smallest number of layers consistent with them; strands that never conflict share a layer, so the layer count follows from the ordering rather than from the number of instances. Each instance is then placed at a height respecting its order that keeps strands from interpenetrating at their measured radii, with the stack compacted so the film is as thin as those constraints allow.

S2.5 Configuration

Every tunable quantity of both stages (the cost weights and gates above, the core and window sizes, the abstention threshold in score units, and the exact-solver size limit) is carried in a single released parameter file, grouped by the stage it belongs to, with the 2-D grouping section checked against the frozen record that produced the published numbers. Table S1 gives that file's 2-D half; Table S2 gives its depth half, so both are shown rather than asserted. Every value in either table, and every parameter this section quotes in passing, is read from the released file when this document is built.

S3 Measures

Every measure this paper reports is defined here, with what it rewards and its blind spot; none reads the rendered width of an instance.

The common-fragment set, shared by the first five measures below. The mask is skeletonised, spurs of at most 3 px are pruned, skeleton pixels with three or more 8-neighbours

Table S2. The depth stage's own entries in the released parameter file. The abstention threshold is in score units, not a band around a probability, and the core is flat: no per-rod radius enters the evidence path.

Quantity	Value	What it sets
core (px)	7.5	half-length of the shared crossing arc; flat at every crossing, so no per-rod radius enters the evidence path
window (px)	22	arclength sampled either side of a crossing for evidence
min. flank	6	fewer usable flank samples and the crossing abstains
abstain $\lvert s \rvert$	0.4	abstention threshold, in score units, not a probability band
exact limit	14	components to this many nodes are ordered exactly, larger ones greedily

are deleted, the remaining 8-connected pieces are labelled, and pieces shorter than 6 px are discarded, so one fragment is one unambiguous arm between crossings. The partition reads neither the reference nor any prediction, so every method compared is cut into the same fragments. Each then takes a reference and a predicted identity by one rule, applied to both sides alike: the fragment is assigned to the instance whose mask overlaps the largest number of its pixels, the vote counted against each instance's full mask rather than a flattened label image, so an instance that overlaps another keeps its pixels, provided that the winning overlap covers at least 30 % of the fragment; failing that, it falls back to the most frequent nearest instance label within 6 px of the fragment, read from a deterministically flattened field; failing both, it stays unassigned and is scored as its own singleton cluster, on the reference side and the prediction side equally, so every fragment enters the scored partition. Ties at any step resolve to the lowest instance id, and the flattening order is fixed, so the assignment is deterministic. Because junction pixels are deleted first, no fragment contains one: none of these measures can see whether an emitted crossing pixel is given one owner or two, only whether the arms meeting there are grouped as the reference groups them.

Sensitivity to the scorer's constants. Because the four constants above are shared with the method (the article's metrics section), the stored predictions of all four methods were re-scored under one-at-a-time perturbations of each constant, with the grid frozen in writing before any result was computed: spur pruning 2–4 px, minimum fragment length 4–8 px, overlap threshold 0.20–0.50, fallback distance 3–9 px. Nothing was re-run and no configuration was changed in response. Table S3 reports PLECTA's mean F_1 and its plain paired margins over each comparator at every perturbation, over the 50 degraded comparison scenes. Across the 10 combinations PLECTA's mean F_1 spans 0.71–0.86, and the smallest paired margin over any comparator under any perturbation is 0.11. The level moves only with the minimum fragment length, and in the same direction for every method, because admitting shorter fragments adds hard, low-evidence units for everyone; the other three constants move the mean by less than 0.002, and no perturbation changes the ordering of the four methods.

Common-fragment pairwise $\mathbf{F_1}$, the primary endpoint. A pair of fragments is a true positive when it shares both identities, a false positive when the prediction joins fragments of different reference strands, a false negative when it separates fragments of one:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad F_1 = \frac{2PR}{P + R}. \tag{S2}$$

A clustering measure of the Rand family [1–3], permutation-invariant in the instance labels; it rewards holding a strand together across every crossing and gap it passes. Blind spots: it is quadratic in fragmentation, so a strand arriving in n pieces carries $\binom{n}{2}$ pairs and a broken mask

Table S3. Scorer-constant sensitivity. Stored predictions re-scored under one-at-a-time perturbations of the four common-fragment constants; Δ columns are plain paired mean margins, PLECTA minus comparator, over the scenes both methods completed. Sensitivity reporting only; the published constants were not changed in response.

Scorer constants	PLECTA F_1	Δ greedy	Δ DNAi	Δ GraFT
published constants	0.82	0.15	0.48	0.34
spur prune 2 px	0.82	0.15	0.48	0.34
spur prune 4 px	0.82	0.15	0.48	0.34
min fragment 4 px	0.71	0.11	0.39	0.27
min fragment 8 px	0.86	0.15	0.51	0.36
overlap threshold 0.20	0.82	0.15	0.48	0.34
overlap threshold 0.40	0.82	0.15	0.48	0.34
overlap threshold 0.50	0.82	0.15	0.48	0.34
fallback distance 3 px	0.82	0.15	0.48	0.34
fallback distance 9 px	0.82	0.15	0.48	0.34

is charged many times for one missed join, and it weights long, much-crossed strands far above short ones.

Adjusted Rand index [4] counts the same pairs corrected for chance, which matters because with a hundred strands per scene almost every pair genuinely belongs to different strands and the uncorrected index sits near 1 for anything. Blind spot: redundancy, since over the comparison set it never departs from pairwise F_1 by more than 0.02, so it confirms rather than adds.

Variation of information [5], in bits, lower better, is the conditional-entropy distance between the partitions: $\mathrm{VI}_{\mathrm{split}} = H(\text{prediction} \mid \text{reference})$ is paid for cutting one strand into several, $\mathrm{VI}_{\mathrm{merge}} = H(\text{reference} \mid \text{prediction})$ for fusing several into one, and reporting both is what separates a method that divides from one that fuses. Blind spot: each component alone is degenerate, since a method merging everything never splits anything and scores a perfect zero on the split component, so neither is ever quoted singly.

Detection $\mathbf{F_1}$, the measure DNAi reports, matches predicted to reference *objects* greedily by highest intersection over union, stopping below 0.1, DNAi's own shipped threshold, with precision over predicted and recall over reference objects. It uses a symmetric 2 px placement tolerance, under which a reference pixel counts as found when the prediction passes within it and vice versa, the intersection being the smaller of the two counts. The tolerance is necessary: two independently read one-pixel centrelines two pixels apart intersect in *nothing*, so a strict overlap reports placement rather than shape, a problem the thin-structure benchmark FISBe states as a design rationale [6]. What keeps that tolerance a correction rather than an inflation is measurable, and the real fields supply the control: on their manual-derived axes, where the input mask is a skeleton of the annotation itself and nothing is displaced, the tolerance moves detection F_1 from 0.93 to 0.93 on B58_110, against 0.18 rising to 0.60 on the nnU-Net axis of the same field. It pays where two readings disagree in placement and nowhere else. Counting objects makes it the most forgiving reading of a fragmented mask and the least discriminative measure here. Blind spot: an arm swap, which it scores 1.000 where the pairwise score gives 0.000.

Crossing Fidelity states the contribution directly. At every junction node the reference induces a partition of the incident arms, two together exactly when the reference gives them the same strand, and so does the prediction; Crossing Fidelity is the fraction of crossings at which the two partitions are identical. Nodes are the junction clusters of the same pruned skeleton and a node's arms are the fragments 8-adjacent to it. Only nodes whose arms carry at least two distinct reference strands are scored, since a node where every arm belongs to one strand is a skeletonisation artefact and scoring it would reward doing nothing. It is an in-house diagnostic, not an established metric, though it follows measures that score correctness at branch points rather than per pixel [7, 8]. Blind spots, both of which govern its use: it is far less discriminative than the pairwise score, so it says what was solved and is never a ranking axis; and crossings carry no information about *gaps*, so it is silent on the cost the pairwise score pays for bridging them. Sharing its skeleton and assignment with the pairwise score makes it a sharper view of the same evidence rather than independent confirmation, and it is always printed beside its all-unpaired baseline.

Join $\mathbf{F_1}$ counts decisions rather than pairs, for the real fields where the two mask sources fragment differently. A *decision* is a pair of distinct fragments whose endpoints come within 85 px, the gap this method will bridge; reference and prediction each say join or not, and precision, recall and their harmonic mean are taken over decisions. Being linear rather than quadratic in fragmentation, and conditioned on the mask, it asks two mask sources comparably hard questions, and splitting and merging appear separately as low recall and low precision. Blind spot: a fragment pair the mask never brought within reach counts against nobody, so the measure is insensitive to a strand the mask lost entirely, and it is always quoted with the number of decisions each mask posed.

Completeness, correctness and quality characterise the input *mask*: no reconstruction is opened to compute them and no method can change them. Reference axis and test mask are both skeletonised and each tested against the other dilated by the tolerance. Completeness is tolerant recall of the reference axis, correctness tolerant precision of the test axis, and quality the tolerant Jaccard index $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$ with TP the smaller of the two matched counts, so quality cannot exceed either component. Quality is also GraFT's published Jaccard index in its intended reading. Blind spot: the tolerance carries the measure, since at zero tolerance it reports raster placement rather than geometry, so it is never quoted without one.

Shared-pixel precision and recall are precision and recall over the pixels the reference gives to more than one instance, computed on the raw instance masks. They exist because the pairwise measure discards junction pixels and is insensitive to whether a crossing pixel was marked as jointly owned at all: only a method emitting overlapping layers can score above zero on the recall, and one emitting a hard partition scores exactly zero by construction. Blind spot: neither component distinguishes a pixel shared for the right reason from one shared because a chain was painted with a whole junction cluster.

The runtime scaling exponent is the slope of an ordinary least-squares fit of log seconds on log foreground pixels, so an exponent b means cost grows as the b-th power of the amount of ink. It is reported instead of absolute time because these are independently optimised codebases, where absolute time measures implementation as much as algorithm. Blind spot: it describes a median trend and is silent on the tail, so it is quoted beside the per-density medians and the count of scenes stopped at the budget. Dropping those scenes biases the exponent downwards for any method whose slowest ones did not finish, which is why GraFT's is a lower bound.

Two measures were computed and are not endpoints. GraFT's *filament matched coverage* asks whether each reference strand was matched one-to-one to some detection, and PLECTA ranks *below* both comparators on it: at the densest stratum of GraFT's own generator, 0.80 against 1.00 and 0.90. It is not an endpoint because it counts existence rather than correctness: a method emitting more objects than the scene contains has a spare for every strand, each method's matched coverage equals the smaller of one and its emission ratio to within 0.16, and requiring a detection to cover half the axis it claims restores the order. *Centreline Dice* [9] was tried as an alternative matching criterion and not adopted: it tests each side's skeleton against the other's *volume*, and a centreline grouping has none, so it degenerates into the placement test the tolerance exists to correct. No clDice result is reported in this paper.

Admitting a reference at all. Every measure above compares a prediction with a reference, so a defective reference is invisible to all of them: it will simply score every method a little lower and change no ordering. A reference is therefore admitted only if it reproduces itself, that is, only if scoring it as though it were a prediction of itself returns a perfect score on every measure. Anything less means the reference disagrees with its own fragment assignment, and the shortfall bounds what any method could score against it. The check is cheap and it is not redundant: it caught a defect that the completeness, correctness and quality triple all passed, because those characterise a mask and this characterises the identities carried on it.

S3.1 Statistical analysis

Held-out means and density strata are accompanied by 95% nonparametric bootstrap intervals (20,000 replicates; seed 20,260,810), resampled with stratification by density; the clean-minus-degraded analysis resamples paired scene differences within density. Per-coverage points in the comparison figures are plain means of ten scenes with no interval claimed. On the real SEM fields the inferential unit is the field, and the 2566 reference crossings entering the crossing measures are a count pooled over six condition-specific evaluations of the three fields, not a sample size. Active-component ablations are one-at-a-time development experiments on all 84 development scenes and do not reuse the held-out set.

S4 Comparator configuration

The article's comparator section states each adaptation and its cost; this is the mechanism behind it, for a reader checking that the comparison was run fairly. **DNAi** clusters junction pixels by single linkage and erases a disk at each cluster. At its released 10 px linkage distance, crossings closer than that chain into one cluster, which at 60 % coverage reaches 89 px across and is replaced by a single small erasure disk, leaving a whole tangle as one instance, which is why its shipped default is not a fair measure of it. Reducing the distance to 2 px on development scenes raised development F_1 and cut variation-of-information merge by about a factor of three, confirming the mechanism; the tuned configuration is the one reported. Two structural choices account for the residual difference: it commits to a pairing at every junction of degree two to four with no cost threshold, so it cannot decline a continuation the way a priced unmatched option can, and it attempts no junction above degree four, resolving 0.02 of the crossings above it. Those are about a tenth of the crossings, but they carry 37.2 % of same-strand fragment pairs on the degraded masks, one such junction severing every pair that spans it. **GraFT**'s errors are divisions rather than merges (variation-of-information split 1.17 against merge 0.39), which follows from a greedy longest-first path cover that cannot revise an early commitment once the scene collapses into one connected graph, as it does above 30 % coverage.

SIFNE enhances the image with an orientation filter transform, thresholds it by Otsu, thins it, and then works from the binary skeleton alone: it marks every pixel whose eight-connected neighbourhood sum reaches three as a crossing, erases each such pixel together with a square neighbourhood, and pairs the resulting fragment termini inside an angular search fan, bidirectionally and one pairing at a time [10]. Because everything after the thinning reads a binary axis, its own re-entry point is the thinned array, and the mask is injected there; every stage downstream is its released source unmodified. Only its graphical callbacks are re-expressed, so that it can run unattended, and its manual-correction stage is never invoked, since every comparator here runs without a human in the loop. One shipped default is degenerate at these densities and is re-selected on development scenes.

The mechanism is measured rather than inferred. Its erasure neighbourhood is 7 pixels square, applied at every pixel whose eight-connected neighbourhood sum reaches three, so the erasures multiply with the crossings: on scenes at 20, 40 and 60 % coverage the surviving mask is 51, 44 and 31 % of what was handed in, and the evidence is gone before any pairing is attempted. On the sparse single-molecule fields the method was built for this costs almost nothing. Selection was staged and both grids were fixed before either ran: first the three settings that govern how much mask survives, then the linkage gates at the winner of the first. Only the ten development scenes were read, never the comparison set. The chosen configuration reduces the erasure to 5 pixels square and the search radius to 35 pixels, raising development F_1 from 0.537 to 0.749 and crossing fidelity from 0.502, below the all-unpaired baseline, to 0.831 well above it. On the comparison set the same change is worth 0.55 to 0.71, and both are reported.

Two further points bear on reading its column. The search-fan angle turns out to be inert at the selected angular gate, because the gate is the stricter of the two tests for every fan wider than sixty degrees and therefore binds first; this was checked against the source rather than assumed from three identical sweep rows. And the neighbourhood radius that its front end uses for orientation filtering is *not* inert once the mask is injected, because the same value sets the arclength over which every tip direction is estimated, which is the whole evidence its pairing sees.

Its crossing excision is not a scoring artefact. It deletes each crossing pixel and its neighbourhood and never restores them, but the common-fragment scorer also deletes junction pixels before labelling, and scores only whether the arms meeting there are grouped as the reference groups them. SIFNE links across the excised crossing, so it makes exactly the decision the endpoint measures. It does score zero on the shared-pixel measures, which is a structural property of a method that assigns no crossing pixel to anything.

Basu et al. is the one comparator with no released implementation: the only code its authors published covers the earlier centreline-localisation paper, at a departmental address that no longer resolves and that no archive captured. What is measured here is therefore a reimplementation of the published principle rather than the authors' program, and it is reported on the same footing as the greedy continuation baseline, a transparent floor and never an external result. Only the geometric half of the method is reimplemented: a minimum spanning tree over the centreline pixels, deletion of every bifurcation node, and recombination of two- and three-segment groups scored by the curvature energy of a fitted smoothing cubic spline [11] less a bond energy, chosen by a set-partitioning integer program that is re-solved until the segment set stops changing. That half reads geometry alone and so accepts an axis mask directly. The other half, an intensity filter bank and a reverse diffusion that converges particles onto brightness crests, is not reimplemented, because scoring our own front end would measure our filter bank rather than their idea; the consequence is that this arm speaks to the recombination logic only.

Its two user parameters, the maximum linking distance and the maximum tangent mismatch, were selected on the same ten development scenes and by the same criterion used for the others. Their published values, fixed in that paper for its own simulated benchmark, score 16 and 10 here only after selection; at the published pair the reimplementation scores 0.076 on these masks, because a two-pixel linking distance admits almost no recombination on a mask whose gaps are wider than that. Both figures are recorded. Because the paper underdetermines two further quantities that change the result, the smoothing of the fitted spline and the quadrature of the bond energy, each was resolved in whichever direction favours the method, and the direction is recorded alongside the choice.

The comparison is read on crossing fidelity as well as on the pairwise score, and the two separate cleanly here: the recombination decides junctions almost as well as the exact matcher does, and loses the pairwise margin on the assembly built from those decisions, because a pairwise score over fragments is quadratic in fragmentation while a junction count is not.

S4.1 Margins the main text states without their intervals

The article's comparator section gives the crossing-resolution rates and the restricted DNAi comparison in summary. The paired figures are here.

Crossing Fidelity, paired against PLECTA over the degraded set: +0.065 [+0.047, +0.083] over the greedy baseline, +0.123 [+0.085, +0.165] over GraFT and +0.185 [+0.158, +0.211] over DNAi. On clean axes PLECTA resolves 0.83. Leaving every arm unpaired resolves 0.09 of these crossings, which is the all-unpaired baseline the rate is read against: a deterministic do-nothing policy rather than a chance level.

Two things about that column belong with it. It is far less discriminative than the pairwise score, spanning 0.26 between best and worst method at 60 % coverage where pairwise F_1 spans 0.54, so it should be read as a count of what was solved and not as a ranking. And every method is poor at the busiest nodes: pooled over the 3638 reference crossings of the degraded set, PLECTA partitions 0.81 exactly, but above degree four it resolves only 0.41, and no comparator does better there.

The restricted DNAi comparison, which scores only the evidence a degree-limited, gap-blind method can reach, moves the clean-axis difference from +0.414 to +0.284 [+0.262, +0.307], still favouring PLECTA in 49 of 50 scenes.

S5 Depth measures

The measures of Section S3 score which centreline fragments belong to one instance. None of them can see depth: two reconstructions that agree on every instance but disagree about which strand lies on top at every crossing are indistinguishable to all of them. The 2.5-D stage is therefore scored by its own family, defined here.

Crossing order accuracy. Each reference crossing names the instance physically on top; the reconstruction either names one or abstains. Three quantities are reported and none is quoted alone: *decided only*, the fraction of committed decisions that are correct, which by itself is not evidence because abstaining on every hard crossing scores 1.000; *all crossings*, which counts an abstention as an error and is the headline; and the *abstention rate* beside both. Chance is exactly 0.5, the decision being binary and symmetric, so only the margin above it carries information. A predicted crossing is credited only when it involves the same instance pair as a reference crossing and lies within the placement tolerance, so inventing crossings cannot help.

Depth order over instance pairs. Crossing accuracy is local. The global counterpart asks, for two instances, whether the reconstruction places the deeper one deeper, and is reported twice. Over *crossing pairs* it measures the stack where evidence exists. Over *all pairs* it includes instances that never cross and are therefore constrained by no evidence at all, whose relative depth is fixed by the solver's tie-break rather than by the image. The two are never pooled, and the fraction of pairs sharing a component is reported beside them so the second can be read against what it could possibly measure. Both are invariant to monotone rescaling of depth, so they describe the arrangement rather than its units. Kendall's τ over the same pairs is $2a - 1$ for accuracy a and is not reported separately.

Layer structure. Layers are an ordinal quotient of depth: a reconstruction can order every pair correctly yet cut the film into a different number of layers. Layer agreement, the fraction of instances placed in the reference layer, is therefore reported separately.

S6 Evaluation split and seeds

The article states only that every parameter was fixed on a separate development set and never on the scenes it reports. This is the mechanism.

Development used 84 scenes spanning 20–60 % coverage. Every parameter of the reconstruction, every gate and price, and the round schedule were settled there, as was each comparator's parameter translation, so that no method was tuned on the scenes it is scored on and all were tuned on the same ones.

The evaluation set is 128 scenes, 32 at each of 20, 30, 40 and 60 % coverage, generated from seed blocks disjoint from development. Disjointness is not asserted but checked: a compact manifest records every seed block, and an executable verifier confirms the counts, the disjointness and the scene metadata it can reach. A further 125-scene locked set exists and *remains unevaluated*; it is reserved against the possibility that the evaluation set above is eventually read too often to stay honest.

Two qualifications follow from this design and are stated wherever they bear on a number. First, the evaluation scenes are an independent *sample* but not an independent *distribution*: they are fresh draws from the generator that produced the development scenes, so their intervals describe sampling within that generator and not transfer to another, which is why the article re-tests its ranking on a generator built by someone else and on real SEM masks. The generator's own shape parameters were also swept rather than left at their defaults. Taking its bending stiffness from 50 to 150 % of default, which moves the median centreline radius from 975 to 308 px, moved the level means of F_1 by at most 0.05 over 4 series, against a within-level scene-to-scene scatter of 0.05; taking the curve wavelength from 10 to 40 px moved them by at most 0.08, and the one series of 4 whose spread exceeded its own noise is non-monotonic with its minimum at the default, which has no mechanism behind it. Both sweeps are development-scene measurements, 10 scenes per level, reported as plain means. Second, any figure computed on the development scenes, the component ablations and the width-sensitivity sweep among them, is labelled as such and is not a held-out effect estimate.

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